

PRACTICAL BOILER WATER TREATMENT HANDBOOK

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Practical Boiler Water Treatment Handbook

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PREFACE

This book is an outcome of my lengthy experience and the burning desire to write books of this type. Starting from plant managers to plant operators indulging in boiler water treatment, they have an umpteen number of unanswered questions. A vast literature is available on this subject, but most of them are descriptive and provide less practical information. As pointed out by George S.Solt and Chris B.Shirley in the preface of their book, "An Engineer's Guide to Water Treatment", "most of the books are written by the experts for the experts". This is true with the case of boiler water treatment books also and therefore most of the technical persons are unaware of the several problems that occur during treatment eventhough they read many books. They carryout treatment with only partial understanding of the basics and intricacies of the treatment. I tried to provide much more practical information on the subject and to offer answers for the questions that have not been cleared hitherto, which sets this book as a different one from others of this type.

This book is divided into nine parts for easy understanding of the subject. Part – I, "Boiler Basics" provides information on the fundamentals of boiler, boiler components and types of boilers. In Part – II, "Boiler Water Troubles", are dealt exhaustively the innumerable troubles caused by the impurities present in water. A thorough understanding of the causes and effects of boiler water troubles would help to eliminate them in the early stages itself. In Part – III, "Water Quality Requirements and Treatment Programs", the quality requirements of feed water and boiler water as prescribed by the American Society of Mechanical Engineers, and recommended by James W.McCoy in his book "The Chemical Treatment of Boiler Water" are given. This is followed by a brief discussion on external and internal treatments and guidelines on treatment programs. In Part – IV, "External Treatment" and Part – V, "Internal Treatment" all the treatment methods are dealt exhaustively.

It is extremely important that the technical persons must have a thorough knowledge on the calculations of blowdown, cycles of concentration and dosage of chemicals. The methods of calculation of such parameters are provided along with worked out problems under Part – VI, "Boiler Water Treatment – Important Calculations". To cater more information on boiler start up and allied subjects, Part – VII, "Boiler Startup, Cleaning, Layup and Maintenance" is included. In Part – VIII, "Chemicals Handling, Solution Preparation and Feeders" are given the methods of chemical handling and storage, methods of preparation of solutions and different types of chemical feeders. Part – IX, "Analysis of Water and Steam" provides analytical procedures for testing feed water and boiler water and for the measurement of steam purity. In the appendix are provided a number of tables which are useful tools to simplify calculations and computations.

This book would be extremely useful to Plant Managers, Water Treatment Personnel, Water Chemists, Boiler Operators and to Water Treatment Consultants and Firms engaged in

(ii)

water treatment as well as for Boiler Manufacturers. This book would serve as a handy reference and ready reckoner to each wet processing industry and power stations and would help to choose the right type of treatment, right type of chemical and to reduce the cost on water treatment significantly. This book would also serve as a ready reference to students of Industrial Engineering, Mechanical Engineering, Chemical Engineering, Industrial Chemistry, Applied Chemistry and to various other faculties of Colleges and Universities.

The usefulness of the book is enhanced by some figures and tables which are reproduced from certain books and journals. Many of the publishers and manufacturers of water treatment equipments and organizations have been kind enough to permit me to reproduce such figures and tables from their publications. I am ever grateful to all of them.

In the preparation of this book, considerable assistance has been derived from a number of my friends, I am grateful to them for their interest, encouragement and assistance. Especially I wish to acknowledge the help rendered by Mr.S.Muruganandam and Mr.Manoharan. I also acknowledge Ms.S.Punitha for the excellent typesetting of this book. Lastly it is a pleasure to extend my sincere thanks, to my wife M.Sivabagiam for her patience and continued encouragement, and to my son Mr.M.Sakthikumar who has designed the cover excellently, amidst his busy schedule.

I am immensely pleased to express my gratitude and sincere thanks to M/s. Chemical Publishing Company, for their keen interest shown towards the publication of this book.

- N. MANIVASAKAM

This book is divided into the following parts

PART – I.	BOILER BASICS
PART – II.	BOILER WATER TROUBLES
PART – III.	WATER QUALITY REQUIREMENTS AND TREATMENT PROGRAMS
PART – IV.	EXTERNAL TREATMENT
PART – V.	INTERNAL TREATMENT
PART – VI.	BOILER WATER TREATMENT – IMPORTANT CALCULATIONS
PART – VII.	BOILER STARTUP, CLEANING, LAYUP AND MAINTENANCE
PART – VIII.	CHEMICALS HANDLING, SOLUTION PREPARATION AND FEEDERS
PART – IX.	ANALYSIS OF WATER AND STEAM

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PART I

BOILER BASICS

Chapter - 1

BOILER – AN INTRODUCTION

Water is the universal solvent and is invariably used for a wide variety of domestic and industrial purposes. The major industrial uses of water are steam generation, cooling, washing, conveying of materials etc., Above all, it is the major raw material which will become an ingredient in the finished product.

Of the many uses of water, steam production is considered as the most important as it is a means of power generation and of several industrial uses. Steam is used to drive turbines and for heating, and to operate pumps, compressors and reciprocating engines. Most chemical process industries require an uninterrupted supply of steam to be used for varied purposes.

Steam is the vapor phase of water. While many liquids can be vaporized and utilized, steam is the preferred one as it can hold very large quantities of heat within it, the property which the vapors of other liquids do not possess. A kilogram of steam carries more energy than any other vaporized liquid. When water is converted to steam, its volume increases by about 1600 times and produces a very high force similar to that of an explosive and hence is used to drive turbines, pumps and compressors. This apart, water is available in plenty than any other material. Besides, handling of water and steam is much easier when compared to handling of other liquids and their vapors.

Steam is generated from water by the common equipment 'Boiler'. A boiler is an enclosed vessel in which heat is transferred continuously from an external heat source to water until it is converted to hot water or steam. The hot water or steam under pressure is then used for transferring the heat for industrial processes or for power generation. The pressure vessel consists of a furnace (heating source), steam drum and mud drum, riser tubes, downcomer tubes, flue gas tubes, steam headers along with necessary accessories and fittings such as safety valves, blow off valves and pressure and temperature gages. This apart, some auxiliary equipments such as economizers, superheaters etc., are also incorporated in a steam generation system. All these components are described separately in later pages.

Boilers can burn coal, oil, natural gas, biomass and other fuels. During combustion, oxygen reacts with the fuel to produce a flame and hot combustion gases. These combustion gases transfer their heat to water in the boiler through radiation, conduction and convection. The relative percentage of each is dependent upon the type of boiler, heat transfer surface and fuel. A boiler is designed in such a way to absorb the maximum amount of heat released in the process of combustion.

Boilers are manufactured in many different sizes and configurations depending on the characteristics of the fuel, the required output, and on the emission controls. Some boilers are only capable of producing hot water while others are designed to produce steam.

There are a few fundamental types of boilers and many variations of each type. Boilers are generally classified according to the operating pressures and operational design. They are also classified based on certain other criteria too. All types of boilers are dealt exhaustively in Chapter – 2.

STEAM GENERATION SYSTEM

It should be noted that boiler is a pressure vessel transforming water to steam while 'steam generation system' is a combination of boiler along with preboiler system and postboiler system. A schematic diagram of a steam generation system is shown in Fig 1 – 1.

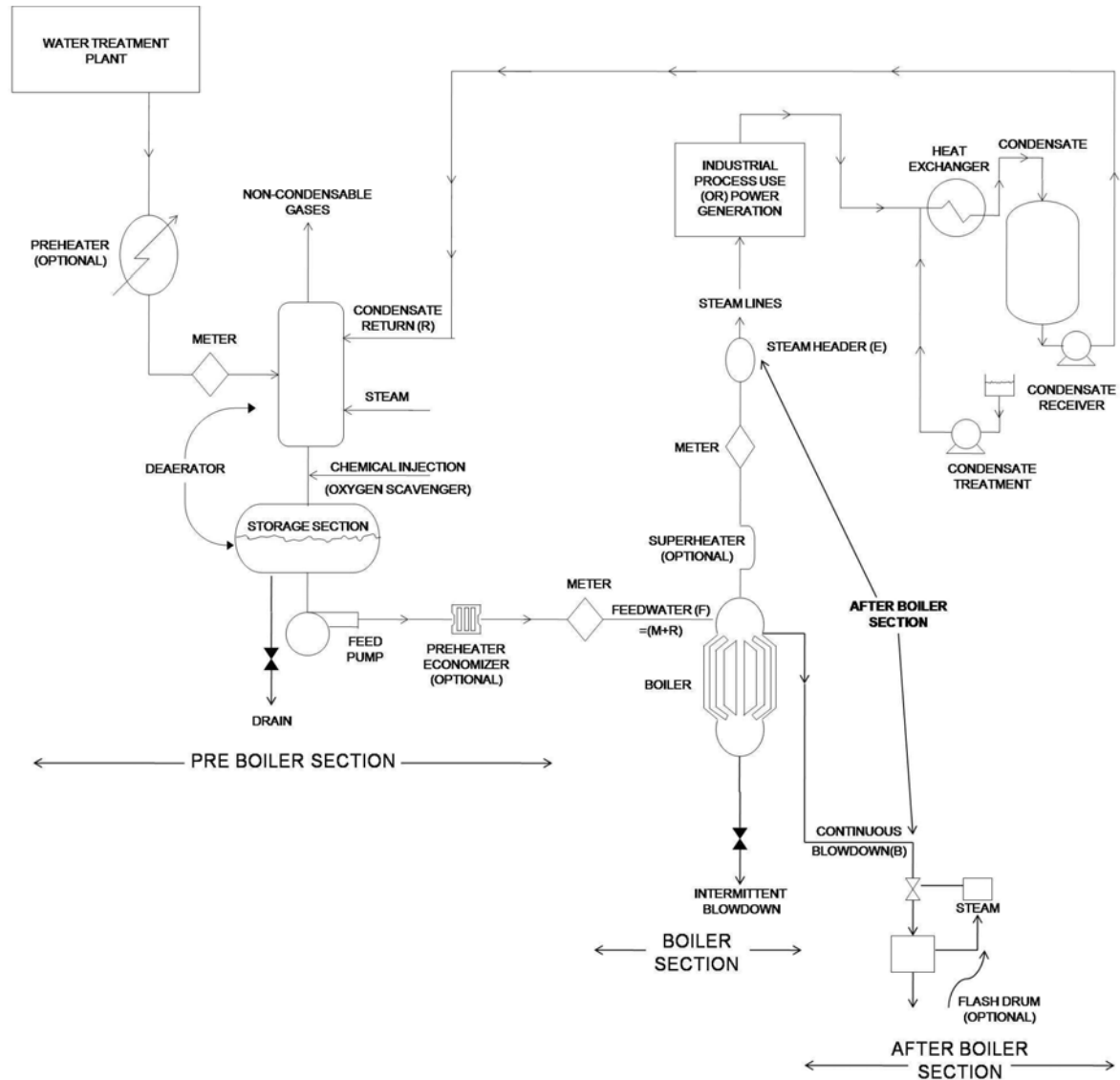


Fig 1 – 1. Schematic Diagram of a Steam Generation System

Pre-boiler Section

The preboiler section of a steam boiler comprises of the boiler system's structural components that hold, move and treat the water before the water enters the boiler. The preboiler system components include water treatment system, feed water pre-heater, deaerator to remove dissolved oxygen (and other non-condensable gases), economizer, and feed water pumps. All these components are discussed in the following section. The pre-boiler section may include a storage tank for the treated makeup water.

Boiler Section

This is exclusively boiler which generates steam. The components of boiler includes steam drum, mud drum, riser tubes, downcomers, and superheaters which are discussed in the following section.

After - boiler Section

The after boiler, or post boiler section comprises of all structural components of the boiler system that hold, move and process the steam and water downstream from the actual boiler. The after-boiler components include steam headers, desuperheaters, condensate system and steam traps etc., which are described in the following section.

BOILER COMPONENTS AND AUXILIARY EQUIPMENTS

Feed Water Pre Heaters

Boiler feed water is usually preheated by the waste heat extracted from spent steam. Closed tube and shell type heat exchangers are used for feed water preheating. These closed feed water heaters generally make use of turbine exhaust steam. The feed water is preheated before it is passed on to deaerators or hot water boilers. In the heat exchanger, feed water is passed in the tubes (inside) and steam on the shell side (outside).

Deaerators

Feed water usually contains objectionable amounts of dissolved oxygen which is highly corrosive. In order to prevent corrosion by dissolved oxygen, it has to be removed completely. Deaeration is the process by which oxygen and other gases are removed from boiler feed water prior to its introduction to a boiler. Deaeration is accomplished by the equipment termed as 'deaerator'. Deaerators function on the principle that solubility of oxygen decreases as the temperature is raised. This is done by passing a stream of steam through the feed water. Deaerators are generally a combination of spray and tray type. A separate chapter is devoted for deaeration and deaerators. Refer Chapter - 38.

Steam Drum and Mud Drum

A boiler system consists of a steam drum and mud drum. The steam drum is the upper drum of a water tube boiler into which feed water is added. Chemicals used for internal boiler water treatment are also

6 Boiler – An Introduction

added to water in the steam drum. Steam drum is the place where separation of water and steam occurs. Steam drums are equipped with mechanical separators to ensure that the steam leaving the boiler does not contain water, solids or other impurities. The present day trend is to use combination of baffles, cyclone-type separators, corrugated scrubbers, and perforated plates for effective separation of steam and water. Steam from the drum may be discharged directly to the steam header or to the superheater to generate superheated steam. The steam drum also contains provisions for continuous blow down.

The steam outlet line normally takes off from this drum to a lower drum by a set of riser and downcomer tubes. The lower drum, called the mud drum is a tank at the bottom of the boiler that equalizes distribution of the water to the generating tubes and collects solids such as salts formed from hardness and silica or corrosion products carried over into the boiler. The mud drum is normally provided with a blowdown valve for periodic blowdown of sediment collecting in the bottom of the drum (See Fig. 2 -5).

Steam Headers

The steam generated by the boiler reaches steam header from where it is being supplied to steam distribution system. (See Fig. 1-1). The steam is either used to generate electricity by passing on to turbine section or consumed for industrial processes. The used steam is then condensed and subsequently reused for steam production.

Economizer

An economizer is a tube bundle placed in the stack of the boiler for preheating feed water. Economizers are used to recover heat from the boiler flue gases and thereby increase boiler efficiency. In effect an economizer is a water heater designed to preheat boiler feed water and deliver the water to the boiler at a high temperature, thus saving energy. By recovering waste heat, an economizer can often reduce fuel requirement by 5% to 10%. Not all boilers have economizers. Usually they are found only on water tube boilers using fossil fuels, that too in high pressure steam plants. Adding an economizer to an existing high pressure boiler installation would increase the capacity of the plant. Boilers that exceed 100 boiler horsepower, operating at pressures exceeding 75 psig or above and those that are significantly loaded all year long are excellent candidates for economizer retrofit. Economizers do not work on most low pressure and HTHW plants.

In power plants that use fossil fuel, improvements in thermo dynamic efficiency are achieved by economizers and air heaters installed in the path of the exit flue gas between the boiler and stack. Economizers transfer heat from the exit gases to the feed water, air heaters transfer some of the residual stack gas heat to the boiler combustion air supply.

An economizer is typically constructed of tubes just like boiler tubes. The tubes can be bare but are usually fitted with fins to increase the heat transfer surface. Bare tubes are used for coal fired boilers, while fin-tubes or extended surfaces are commonly used on gas and oil-fired units. Figure 1 – 2 shows a continuous bare tube economizer. Figure 1-3 illustrates a steel-finned extended surface economizer.

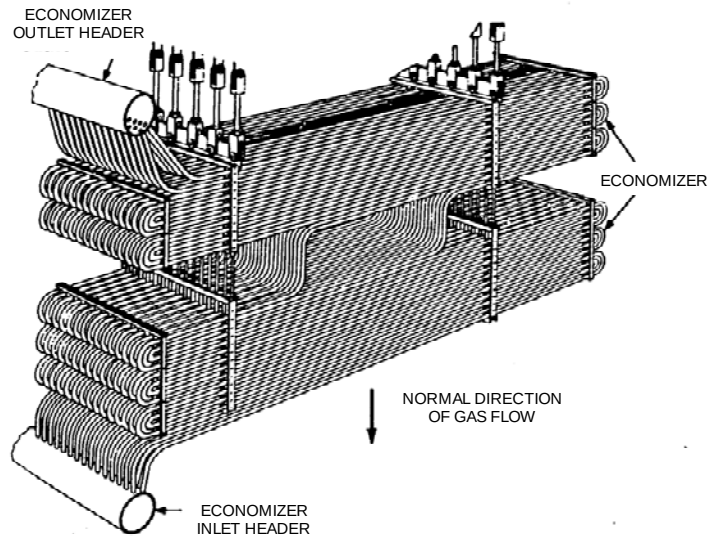


Fig 1 – 2. Bare Tube Economizer*

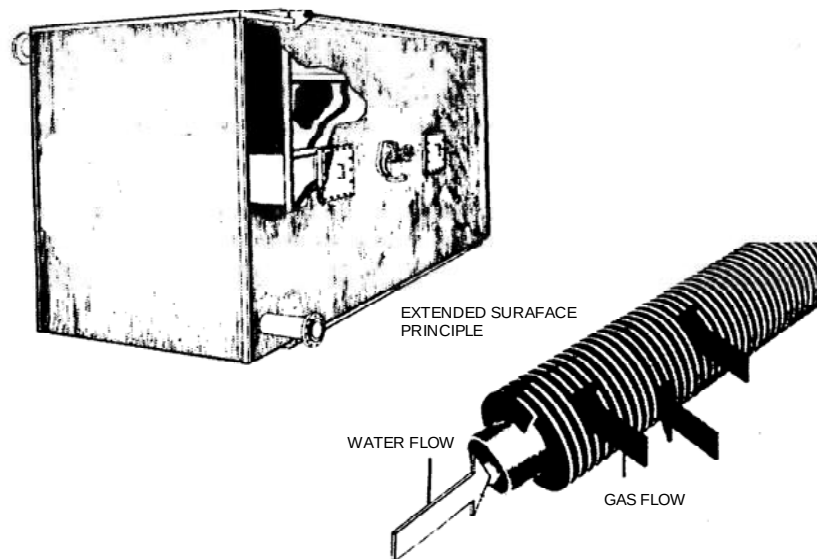


Fig 1 – 3. Extended Surface Economizer*

* Reproduced with the kind permission from the Department of the Army, U.S. Army Corps of Engineers, Alexandria, from their publication, "Central Boiler Plants" (TM 5-650).

Super Heaters

Some industrial processes and turbines require superheated steam which can be obtained by incorporating a super heater in the boiler system. To obtain superheated steam, steam from the steam drum is allowed to pass through superheater where it gets additional heat. The purpose of superheating is to remove the last traces of moisture from the steam and to transfer additional heat to the steam, thus raising the temperature of the steam above its saturation point. In power plants, to achieve the highest possible thermodynamic efficiency, modern fossil fuel power cycles use steam superheaters and reheaters to raise average temperatures of steam flowing through the turbine.

The superheaters can be single bank or multiple banks of tubes either in horizontal or vertical arrangement and located in the convective or radiation zone of the boiler. Fig. 1-4 illustrates a two drum boiler equipped with a superheater, water walls, spreader stoker and economizer.

There are two types of super heaters, namely 1) Drainable Super Heaters and 2) Pendant Super Heaters. Drainable superheaters as the name implies have arrangements to drain the water that accumulates. In pendant superheater, water is not drained, instead the tubes hang down from the heaters into the furnace or flue gas passages.

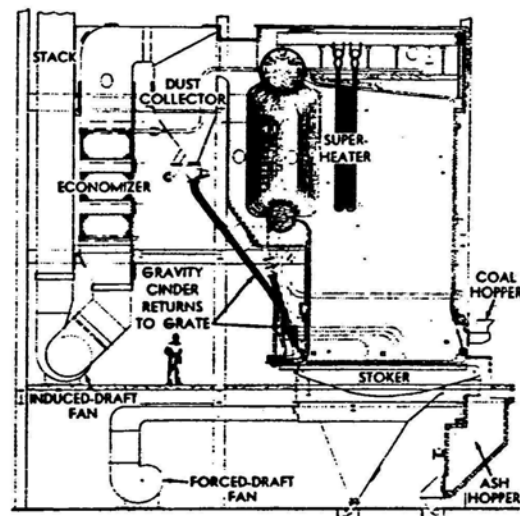


Fig 1 – 4. Super Heater in Two Drum Boiler*

Desuperheaters (Attemperators)

Desuperheater or attemperator is an equipment for reducing and controlling the temperature of superheated steam.

* Reproduced with the kind permission from the Department of the Army, U.S. Army Corps of Engineers, Alexandria, from their publication, "Central Boiler Plants" (TM 5-650).

Normally steam is superheated to increase the cycle efficiency in power generating plants. However superheat is objectionable in process steam, principally because it creates a lower heat transfer rate than does saturated steam. Therefore larger equipment is required if super heated steam is used. Because of this reason, it becomes necessary to desuperheat the steam for process use. Desuperheating is accomplished by injecting a stream of high purity water into the superheated steam. The injection rate is automatically controlled by temperature sensors that modulate the flow of injection water. The injected water evaporates and becomes part of the steam while reducing the temperature (and pressure) of the steam. An incidental advantage is the increase in supply of desuperheated steam. Desuperheaters are usually located at the exit of the superheater, but may also be placed in an intermediate position.

As the concentration of the total dissolved solids in the spray water should be < 2.5 ppm, condensate is the preferred supply. In emergencies, feed water or treated makeup water may be used.

Steam Traps

Steam traps are integral part of any steam distribution system. Steam trap is a device attached to a steam line that automatically drains off any condensate that forms (and other non-condensable gases such as air) without releasing any significant amount of steam. It is a very important mechanical item for maintaining peak efficiency in steam system. Industrial plants normally have a number of steam traps. It should be remembered that no single type of steam trap is ideal for every situation. Four types of steam traps have been developed to meet specific requirements. They are (i) Thermostatic (ii) Float and Thermostatic (c) Disc / Thermodynamic, and (iv) Inverted Bucket types. Adequate care is essential in maintaining steam traps. If steam traps are not maintained, condensate may be retained in the system, severely limiting heat transfer.

BOILER OPERATING PRESSURE

The operating pressure is usually expressed in terms of pounds per square inch (psi). It must be clarified that boiler operators express the pressure indicated by the gauge and hence the precise expression is 'psig' (pounds per square inch gauge). Wherever pressure is indicated simply by psi it is psig only. There is another term 'absolute pressure' which is the gauge pressure plus atmospheric pressure (14.7 psi) and expressed as pounds per square inch absolute (psia). But plant operators are normally interested only in gauge pressure.

Pressure is also expressed in other units. They are (i) Kilogram per square centimeter (Kg/cm^2), (ii) atmosphere (iii) bar and (iv) Pascal (or Newton / meter²). Table 1 -1 provides conversion factors between these units.

BOILER REGULATIONS

Boiler is a pressure vessel, and safety during operation is vital. Keeping in view of the safety, each and every country has promulgated a set of rules and laws with regard to the construction, operation and inspection of boilers. The American Society of Mechanical Engineers has formulated a code for operating boilers. These standards are universal and have been incorporated into the laws of many countries. For more detailed information, readers are directed to refer 'Boiler and Pressure Vessel Code' published by ASME.

TABLE 1 – 1. PRESSURE UNITS AND CONVERSION FACTORS

From	To	Multiply by
psi (or lb/in ²)	Kg / cm ²	0.07031
psi	atmosphere	0.068046
psi	bar	0.068948
psi	Pa (or) N / m ²	6.8927 x 10 ³
Kg / cm ²	psi	14.223
Kg / cm ²	atmosphere	0.9678
Kg / cm ²	bar	0.9804
Kg / cm ²	Pa (or) N / m ²	9.804 x 10 ⁴
atmosphere	psi	14.696
atmosphere	Kg / cm ²	1.03325
atmosphere	bar	1.01295
atmosphere	Pa (or) N / m ²	1.01295 x 10 ⁵
bar	psi	14.5039
bar	Kg / cm ²	1.02
bar	atmosphere	0.9872
bar	Pa (or) N / m ²	1 x 10 ⁵
Pa (or) N / m ²	psi	1.4508 x 10 ⁻⁴
Pa (or) N / m ²	Kg / cm ²	1.020 x 10 ⁻⁵
Pa (or) N / m ²	atmosphere	9.869 x 10 ⁻⁶
Pa (or) N / m ²	bar	1 x 10 ⁻⁵

Pressure is also expressed as MN / m²

1 MN / m² = 1 MPa

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